

Abundance and Biomass of Sand Prawn *Kraussillichirus kraussi* in Richards Bay Harbour.



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Introduction



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Results and Discussion

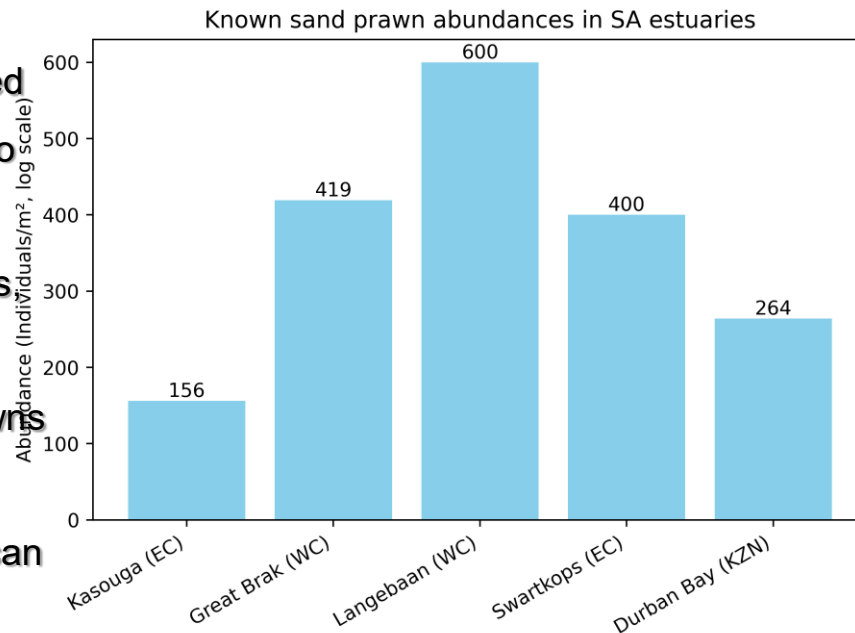
Abundance and biomass.

- Abundance-Biomass Comparison curves showed that Sites A and B had higher biomass relative to abundance, indicating low disturbance.
- Site C exhibited higher abundance than biomass, suggesting severe disturbance.
- This may be attributed to overharvesting of prawns for bait, potential sediment contamination or recreational activities. Site C is situated on Pelican Island, a popular recreational destination.



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Modeling Abundance–Biomass Patterns Using Regression Approaches.

- The relationship between abundance and biomass was examined using linear, polynomial, and Random Forest models.
- There was a positive linear relationship between abundance and biomass.
- More data is required to further analyse (e.g., regression) this relationship.
- In conclusion, abundance and biomass can be used as a quick and cost-effective tool for assessing ecosystem health in Richards Bay Harbour.
- It is recommended that future studies also consider sediment type, contaminants, and predictive models such as Random Forest to improve predictions of macrobenthic abundance, biomass, and community responses.



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